Lab 3: Architectural Justification

Vaccination Cohort & Dose Scheduling System

Architecture Selection: We chose the Microservices Architecture for the Vaccination Cohort & Dose Scheduling System.

Reason 1: Independent Scaling for Divergent Load Patterns

Certificate verification traffic (citizens and third parties scanning QR codes) is far higher-volume and far more latency-sensitive than registration or scheduling traffic, which is dominated by scheduled booking windows. Splitting Certificate Verification into its own service lets it be scaled and optimised independently to meet the sub-150ms NFR without over-provisioning the Registration or Scheduling services, which do not need that throughput.

Reason 2: Fault Isolation During a Public Rollout

A national vaccination rollout cannot allow a fault in one part of the system to take down the rest. If the Notification Service fails or the Registration Service is under heavy load during a new cohort opening, the Dose Scheduling Service and Certificate Verification Service must keep working so citizens already booked can still be vaccinated and existing certificates can still be verified at centers.

Security Advantage

Isolating the Certificate Service means the private signing key used to digitally sign QR certificates lives in exactly one service with a minimal, tightly-scoped API, instead of being reachable from every part of a monolith. The public-facing Certificate Verification Service only ever holds the public verification key and exposes a read-only 'verify' endpoint, so a compromise of the verification path can never expose the ability to forge new certificates.

Performance Benefit

Because the Certificate Verification Service is a small, stateless service focused only on signature checks (with cached public keys), it can respond in well under 150ms and be horizontally scaled with extra lightweight instances during peak travel or entry-checkpoint periods, independent of the heavier Dose Scheduling Service, which performs interval-rule checks and slot-availability queries against the booking data store.